

PSLE MATHEMATICS PRACTICE PAPER 7

Volume & 3D Shapes

Duration: 1 hour

Total Marks: 50

Name: _____

Date: _____

SECTION A: Short Answer Questions (20 marks)

1. Find the volume of a cube with side 3 cm. (2 marks)

2. A rectangular box has length 10 cm, width 6 cm, height 4 cm. Find its volume. (2 marks)

3. Convert 2000 cm³ to liters. (2 marks)

4. A cylinder has radius 5 cm and height 8 cm. Find its volume. (Use $\pi = 3.14$) (2 marks)

5. If 1 liter = 1000 cm³, how many liters is 5000 cm³? (2 marks)

6. Name 5 3D shapes. (2 marks)

7. How many faces, edges, and vertices does a cube have? (3 marks)

8. A sphere has radius 7 cm. Find its volume. (Use $\pi = 22/7$) (3 marks)

SECTION B: Long Answer Questions (30 marks)

9. A fish tank is 60 cm long, 40 cm wide, and 50 cm high. Find its volume in cm^3 and liters. (4 marks)

10. A triangular prism has a triangular base with area 20 cm^2 and length 15 cm. Find its volume. (3 marks)

11. A cylindrical container has radius 10 cm and height 20 cm. Find its volume and surface area. (Use $\pi = 3.14$) (4 marks)

12. Water fills a rectangular tank to a depth of 30 cm. The tank is 80 cm $\times$ 50 cm. Find volume in liters. (3 marks)

13. A pyramid has a square base of side 6 cm and height 10 cm. Find its volume. (3 marks)

14. Compare the volumes of a cube (side 10 cm) and a sphere (radius 10 cm). (4 marks)

15. A cone has radius 6 cm and height 12 cm. Find its volume. (Use $\pi = 3.14$) (4 marks)

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SOLUTIONS

SECTION A SOLUTIONS

Q1: Cube volume

$$V = \text{side}^3 = 3^3 = 27 \text{ cm}^3$$

Q2: Box volume

$$V = l \times w \times h = 10 \times 6 \times 4 = 240 \text{ cm}^3$$

Q3: Convert cm^3 to liters

$$2000 \text{ cm}^3 = 2000 \div 1000 = 2 \text{ liters}$$

Q4: Cylinder volume

$$V = \pi r^2 h = 3.14 \times 5^2 \times 8 = 628 \text{ cm}^3$$

Q5: Convert to liters

$$5000 \text{ cm}^3 = 5000 \div 1000 = 5 \text{ liters}$$

Q6: 5 3D shapes

Cube, Sphere, Cylinder, Cone, Pyramid

Q7: Cube features

Faces = 6, Edges = 12, Vertices = 8

Q8: Sphere volume

$$V = \frac{4}{3}\pi r^3 = \frac{4}{3} \times \frac{22}{7} \times 7^3 = 1437.33 \text{ cm}^3$$

SECTION B SOLUTIONS

Q9: Fish tank

$$V = 60 \times 40 \times 50 = 120,000 \text{ cm}^3 = 120 \text{ liters}$$

Q10: Triangular prism

$$V = \text{base area} \times \text{length} = 20 \times 15 = 300 \text{ cm}^3$$

Q11: Cylinder

$$V = \pi r^2 h = 3.14 \times 10^2 \times 20 = 6280 \text{ cm}^3, \text{ SA} = 2\pi r^2 + 2\pi rh = 628 + 1256 = 1884 \text{ cm}^2$$

Q12: Water volume

$$V = 80 \times 50 \times 30 = 120,000 \text{ cm}^3 = 120 \text{ liters}$$

Q13: Pyramid volume

$$V = \frac{1}{3} \times \text{base area} \times \text{height} = \frac{1}{3} \times 36 \times 10 = 120 \text{ cm}^3$$

Q14: Cube vs Sphere

$$\text{Cube: } 10^3 = 1000 \text{ cm}^3, \text{ Sphere: } \frac{4}{3} \times 3.14 \times 10^3 = 4186.67 \text{ cm}^3$$

Q15: Cone volume

$$V = \frac{1}{3} \times \pi r^2 h = \frac{1}{3} \times 3.14 \times 36 \times 12 = 452.16 \text{ cm}^3$$

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